

CRYSTAL CHENG

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EDUCATION

University of California, Berkeley, Berkeley, CA

Expected Dec 2026

Bachelor of Arts in Computer Science & Applied Mathematics

- Relevant Coursework: Artificial Intelligence, Machine Learning, Statistical Modeling, Linear Algebra, Probability Theory, Discrete Math, Numerical Analysis, Data Structures, Computer Architecture
- Involvements: Society of Women Engineers (Treasurer), Web Development at Berkeley (Developer), Berkeley Project (Developer)

WORK EXPERIENCE

Hyve Solutions | Software Engineer Intern

June 2026 - Present

- Implementing an LLM-powered document processing platform in Python that combines prompt engineering, RAG, REST APIs, and automated output validation to automate technical document analysis for AI infrastructure validation and integration testing

Newrium Foundation Inc | Software Engineer Intern

Sep 2025 - Dec 2025

- Owned development of customer-facing support and FAQ workflows, translating Figma designs into production-ready Next.js features and coordinating frontend integration across a 15-person cross-functional team on a 10-week timeline
- Shipped 15+ reusable React/TypeScript components, cutting implementation overhead and standardizing frontend patterns
- Architected MySQL schemas storing 100+ records and developed a Node.js email automation pipeline with SQL task queues, collaborating on API contract design to enable reliable end-to-end feature integration

Barobo | Software Engineer Intern

May 2025 - Aug 2025

- Blocked unauthorized video sharing across 100% of user traffic by building a single-use link system using AWS CloudFront signed URLs with configurable TTL-based expiration and per-user access control logic
- Added AWS DynamoDB to track user access metadata, scaling the backend to handle 10,000+ video share requests
- Reduced average video load time from ~800ms to ~760ms by adding CloudFront edge caching in front of S3 retrieval, a 5% latency improvement that compounds across all 10,000+ monthly video requests with no backend changes
- Rewrote deployment as reusable Terraform and AWS CLI scripts, bringing provisioning time down from 2 hours to 15 minutes

Opal Mentorship App | Machine Learning Intern

May 2024 - May 2025

- Raised mentor-mentee match accuracy by 12% across cohorts by tuning 6 algorithmic scoring weights over two rounds of iteration
- Built a repeatable evaluation pipeline in AWS Lambda and DynamoDB that scored compatibility across 4 behavioral dimensions, giving the team a reliable and version-controlled basis for weighting decisions at small cohort scale
- Created a JavaScript scoring dashboard visualizing compatibility distributions across cohorts, driving 2 rounds of weight calibration

Swingbeats Research Lab | Frontend Web Developer

Jan 2024 - May 2024

- Redesigned the haptic feedback research lab website in Figma and rebuilt it with React, HTML, and CSS, improving navigation and scaling layout components across desktop, tablet, and mobile device form factors for broader reach
- Refactored the homepage into reusable React components, cutting redundant render logic across 3 page sections

PROJECTS

MBTI Guesser | *Hugging Face Transformers BART-MNLI, DeepFace, OpenCV, Python*

[link](#)

- Developed a free-form MBTI classifier using BART-MNLI zero-shot inference, replacing traditional surveys with a 13-feature multimodal pipeline (text, social graphs, images, and behavioral data) without labeled training data
- Mitigated short-input ENTJ bias by redesigning NLI templates and adding thresholds to eliminate low-confidence false positives
- Fused text/image/numeric signals (65%/25%/10%) using DeepFace and OpenCV, with dynamic reweighting for missing inputs

GreenWays | INRIX Recruiting Hackathon | *Javascript, React, REST APIs, AWS*

- Built a full-stack web application that processed traffic data from the INRIX Traffic API across 500+ roadway segments to identify congestion-driven emission hotspots
- Engineered interactive geospatial visualizations displaying 10,000+ real-time traffic data points with filtering, analytics capabilities
- Integrated third-party APIs and deployed cloud-hosted services on AWS within a 24 hour hackathon development cycle

SKILLS

- Languages: Python, C/C++, Javascript, SQL, Bash, HTML/CSS (Tailwind), MATLAB
- Machine Learning & Frameworks: PyTorch, Sklearn, NumPy, Pandas, React, Bootstrap, Next.js, REST APIs
- Cloud, DevOps, & Tools: AWS (S3, CloudFront, DynamoDB, Lambda), Git/Github, Linux, Docker, Jenkins, Terraform, Postman